Group and Representation Theory

Chapter 5. Roots, Weights and SU(3)

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Introduction

- Now we want to extend our analysis from $SU(2)$ to arbitrary simple Lie algebra, such as $SU(3)$
- The idea of raising/lowering operators generalizes for this setting, by the concepts of **roots** and **weights**
- Furthermore, we examine the structure of the **root space**
- Finally, we apply these concepts to $SU(3)$ **flavor symmetry**, leading us to the famous **eightfold way**

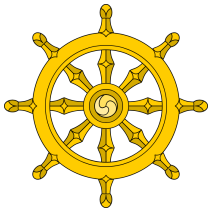


Figure 1: The *Dharmachakra*. The eight spokes indicate the *Nobel Eightfold Path* leading to the truth.

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Cartan Generators

- Given an algebra and generators, we want to diagonalize them as much as possible
- Therefore we think a maximal set of simultaneously diagonalizable generators, called the **Cartan subalgebra**

Cartan Generators

Let D be an irrep of the algebra. A largest possible set of generators $\{H_i\}$ such that

$$H_i = H_i^\dagger, \quad [H_i, H_j] = 0 \quad \forall i, j \tag{1}$$

is called a **Cartan subalgebra**. The generators of the Cartan subalgebra is called the **Cartan generators**.

The Cartan subalgebra forms itself a vector space, thus we can choose **normalization** as

$$\mathrm{tr}(H_i H_j) = k_D \delta_{ij}, \tag{2}$$

where k_D is a normalization constant depending on the rep D

Weights

Rank of the Algebra

The number of independent Cartan generators[a], m, is called the **rank** of the algebra.

[a] The dimension of the vector space spanned by Cartan generators

After diagonalization of Cartan generators, the states of the rep D can be written as $|\mu, x, D\rangle$ where

$$H_i |\mu, x, D\rangle = \mu_i |\mu, x, D\rangle \tag{3}$$

Weights

The eigenvalues μ_i of the Cartan generators are called **weights**, and the m-component vector $\mu = (\mu_1, \cdots, \mu_m)$ is called the **weight vector**.

Weights

e.g. Let $|j, m\rangle$ be a state in spin-j rep of $SU(2)$. Since $SU(2)$ has only one diagonal generator J_3, the rank of $SU(2)$ is 1.

$$J_3 |j, m\rangle = m |j, m\rangle \tag{4}$$

so m is the weight of the Cartan generator J_3

- The weights are **real** since they are eigenvalues of hermitian generators
- We often abbreviate

$$\alpha \cdot \mu \equiv \alpha_i \mu_i, \quad \alpha^2 \equiv \alpha_i \alpha_i, \quad \mu \cdot H \equiv \mu_i H_i \tag{5}$$

The Adjoint Representation

- We've defined the adjoint rep of the algebra as

$$[T_a]_{bc} = -if_{abc} \tag{6}$$

- One remarkable property of the adjoint rep is that the states of the adjoint rep correspond to the generators themselves

$$T_a \leftrightarrow |T_a\rangle\,, \quad \alpha T_a + \beta T_b \leftrightarrow |\alpha T_a + \beta T_b\rangle \tag{7}$$

- In the adjoint rep, the 'action' of the generator T_a on the state $|T_b\rangle$ is the **commutation**

$$T_a |T_b\rangle = |T_c\rangle \langle T_c| T_a |T_b\rangle = [T_a]_{cb} |T_c\rangle = -if_{acb} T_c = if_{abc} |T_c\rangle \tag{8}$$

$$= |if_{abc} T_c\rangle = |[T_a, T_b]\rangle \tag{9}$$

Roots

- Roots are generalization of raising/lowering operators
- Roots determine how the weight vector μ changes when raising/lowering operators are applied

Roots

The **roots** are the weights of the adjoint representation. A vector $\alpha = (\alpha_1, \cdots, \alpha_m)$ is called the **root vector**.

The states corresponding to the Cartan generators have **zero weight** :

$$H_i \left| H_j \right\rangle = \left| [H_i, H_j] \right\rangle = 0 \tag{10}$$

The other states, denoted E_α, have **non-zero weight**

$$H_i \left| E_\alpha \right\rangle = \alpha_i \left| E_\alpha \right\rangle \quad \leftrightarrow \quad [H_i, E_\alpha] = \alpha_i E_\alpha \tag{11}$$

e.g. Let us consider adjoint rep of $SU(2)$

$$J_1^{(\mathrm{ad})} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad J_2^{(\mathrm{ad})} = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad J_3^{(\mathrm{ad})} = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{12}$$

If we diagonalize J_3 with appropriate normalization, in the new basis (J_+, J_3, J_-) we have

$$J_+ = \sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad J_3 = \begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{0} & 0 \\ 0 & 0 & -\mathbf{1} \end{pmatrix}, \quad J_- = \sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \tag{13}$$

J_+, J_- are raising/lowering operators, with **root** ± 1 !

Roots

The generators E_α are **not** Hermitian because taking the adjoint gives

$$[H_i, E_\alpha^\dagger] = -\alpha_i E_\alpha^\dagger \tag{14}$$

So we can take $E_\alpha^\dagger = E_{-\alpha}$ (like $J_+^\dagger = J_-$)

- States corresponding to different roots must be **orthogonal**, since they have different eigenvalues of Cartan generators
- We can choose normalization of the states in the adjoint rep so that

$$\langle E_\alpha | E_\beta \rangle = \lambda^{-1} \operatorname{tr}\left(E_\alpha^\dagger E_\beta \right) = \delta_{\alpha\beta} \tag{15}$$

Raising and Lowering

The $E_{\pm\alpha}$ are the **raising and lowering operators** for the weights, because

$$H_i E_{\pm\alpha} |\mu\rangle = [H_i, E_{\pm\alpha}] |\mu\rangle + \mu_i E_{\pm\alpha} |\mu\rangle = (\mu_i \pm \alpha_i) E_{\pm\alpha} |\mu\rangle \quad (16)$$

So the operator $E_{(\alpha_1,\alpha_2,\cdots,\alpha_m)}$ changes the weight $(\mu_1, \cdots, \mu_m)$ by

$$(\mu_1, \cdots, \mu_m) \to (\mu_1 + \alpha_1, \cdots, \mu_m + \alpha_m) \quad (17)$$

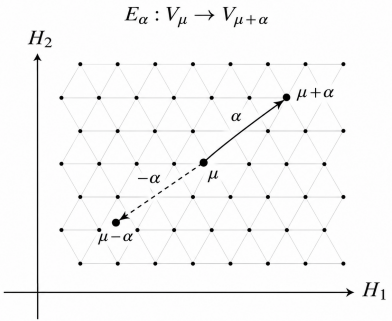


Figure 2: Root vectors can be thought as differences between weight vectors.

SU(2) Substructure

For a non-zero pair of root vectors $\pm\alpha$, there is an $SU(2)$ **subalgebra** of the group with generators

$$E^{\pm} = |\alpha|^{-1} E_{\pm\alpha}, \quad E_3 = |\alpha|^{-2} \alpha \cdot H \tag{18}$$

Check :

$$[E_3, E^{\pm\alpha}] = |\alpha|^{-3} [\alpha \cdot H, E_{\pm\alpha}] = \pm E^{\pm}$$
$$[E^{+}, E^{-}] = \alpha^{-2} [E_{\alpha}, E_{-\alpha}] = |\alpha|^{-2} \alpha \cdot H = E_3$$

So they really satisfy $SU(2)$ commutation relations!

- A state of the full algebra can be decomposed into irreps of the $SU(2)$ subalgebra
- Roots vectors correspond to unique generators
- If α is a root, then no multiple of α other than $\pm\alpha$ is a root

SU(2) Substructure

For example, the diagram below shows weight diagram of the $SU(3)$ adjoint representation $\mathbf{8}$, and its decomposition into $\mathbf{8} = \mathbf{3} \oplus \mathbf{2} \oplus \mathbf{2} \oplus \mathbf{1}$

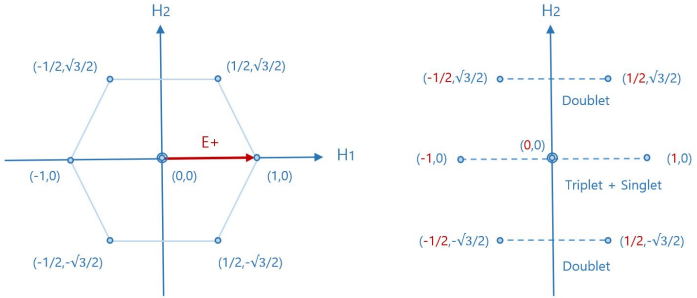


Figure 3: The adjoint representation decomposes into $SU(2)$ multiplets once a root direction is fixed.

The Master Formula

- The most important result of $SU(2)$ substructure is the **master formula**
- The formula strongly constrains structure of the root space, and has a nice geometrical interpretation

For any weight μ, the E_3 value is

$$E_3 \left|\mu, x, D\right\rangle = \frac{\alpha \cdot \mu}{\left|\alpha\right|^2} \left|\mu, x, D\right\rangle \tag{19}$$

But for $SU(2)$, the E_3 value must be an integer or half-integer, so we conclude

$$\frac{2\alpha \cdot \mu}{\left|\alpha\right|^2} \in \mathbb{Z} \tag{20}$$

Now a general state $\left|\mu, x, D\right\rangle$ can be written as a linear combination of states transforming according to definite rep of $SU(2)$ subalgebra :

$$\left|\mu, x, D\right\rangle = \underbrace{a_1 \left|J_1, m_1, \cdots\right\rangle}_{\text{State transforming under spin } J_1 \text{ irrep}} + \underbrace{a_2 \left|J_2, m_2, \cdots\right\rangle}_{\text{State transforming under spin } J_2 \text{ irrep}} + \cdots \tag{21}$$

The Master Formula

Now pick $j = \max(J_1, J_2, \cdots)$, then there exist nonnegative integers p and q such that

$$(E^+)^p |\mu, x, D\rangle \neq 0, \quad (E^+)^{p+1} |\mu, x, D\rangle = 0$$
$$(E^-)^q |\mu, x, D\rangle \neq 0, \quad (E^-)^{q+1} |\mu, x, D\rangle = 0$$

By definition of j, $(E^+)^p |\mu, x, D\rangle$ is the **highest weight state** of the spin j rep, and its weight will be

$$\frac{\alpha \cdot (\mu + p\alpha)}{|\alpha|^2} = \frac{\alpha \cdot \mu}{|\alpha|^2} + p = j \tag{22}$$

$$\frac{\alpha \cdot (\mu - q\alpha)}{|\alpha|^2} = \frac{\alpha \cdot \mu}{|\alpha|^2} - q = -j \tag{23}$$

The Master Formula

Combining these two relations, we get the **master formula** :

$$\boxed{\frac{\alpha \cdot \mu}{|\alpha|^2} = -\frac{1}{2}(p - q)} \tag{24}$$

Now we apply the formula for **roots**

$$\frac{\alpha \cdot \beta}{|\alpha|^2} = -\frac{1}{2}(p - q), \quad \frac{\beta \cdot \alpha}{|\beta|^2} = -\frac{1}{2}(p' - q') \tag{25}$$

Multiplying results gives information about **angles** between **root vectors** α and β!

$$\cos^2 \theta_{\alpha\beta} = \frac{(\alpha \cdot \beta)^2}{|\alpha|^2 |\beta|^2} = \frac{(p - q)(p' - q')}{4} \leq 1 \tag{26}$$

The Master Formula

Since p, q, p', q' are integers, the only possible cases are (Angles 0 and π are not interesting, since no two roots are the same and roots $\pm\alpha$ appear in pair)

$(p-q)(p'-q')$	$\theta_{\alpha\beta}$
0	$\pi/2$
1	$\pi/3$ or $2\pi/3$
2	$\pi/4$ or $3\pi/4$
3	$\pi/6$ or $5\pi/6$

Table 1: Possible 'angles' between roots.

- Actually, the whole structure of the Lie algebra is determined by fundamental objects called '**simple roots**' and their relative lengths and angles
- The structure can be illustrated by **Dynkin diagram**, and ultimately can be classified

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The Gell-Mann Matrices

The standard basis for the $SU(3)$ is called the **Gell-Mann matrices**, which can be viewed as a generalization of Pauli matrices σ^i

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \tag{27}$$

$$\lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{28}$$

- Since $SU(3)$ contains $SU(2)$ as subalgebra, we see that $\lambda_1, \lambda_2, \lambda_3$ contain the Pauli matrices
- There are **2** diagonal matrices, so $SU(3)$ has rank **2**

The Gell-Mann Matrices

The $SU(3)$ generators are conventionally defined by

$$T_a = \frac{\lambda_a}{2} \quad a = 1, \cdots, 8 \tag{29}$$

with normalization

$$\mathrm{tr}(T_a T_b) = \frac{1}{2}\delta_{ab} \tag{30}$$

There are two diagonal matrices T_3 and T_8, so we can form the Cartan subalgebra

$$T_3 = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -1/2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T_8 = \frac{\sqrt{3}}{6}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \tag{31}$$

Roots and Weights of $SU(3)$

Thanks to Gell-Mann, we've found eigenvectors and associated **weights** of $SU(3)$ for free :

$$\begin{pmatrix}1\\0\\0\end{pmatrix} \to (1/2, \sqrt{3}/6), \quad \begin{pmatrix}0\\1\\0\end{pmatrix} \to (-1/2, \sqrt{3}/6) \quad \begin{pmatrix}0\\0\\1\end{pmatrix} \to (0, -\sqrt{3}/3) \tag{32}$$

The roots will be **differences of the weights**, so

$$\frac{1}{\sqrt{2}}(T_1 \pm iT_2) = E_{\pm 1,0} \tag{33}$$

$$\frac{1}{\sqrt{2}}(T_4 \pm iT_5) = E_{\pm 1/2, \pm\sqrt{3}/2} \tag{34}$$

$$\frac{1}{\sqrt{2}}(T_6 \pm iT_7) = E_{\mp 1/2, \pm\sqrt{3}/2} \tag{35}$$

Roots and Weights of $SU(3)$

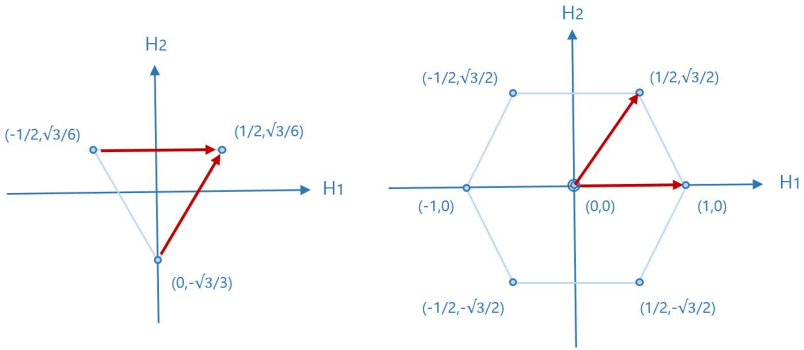


Figure 4: Weight and Root diagrams of $SU(3)$. There are two elements at the center, corresponding to two generators of the Cartan subalgebra.

- **The isospin model**

  Heisenberg(1932) proposed a proton-neutron doublet $N = \begin{pmatrix} p \\ n \end{pmatrix}$, because mass difference was only 0.14%

  $\rightarrow$ Isospin $1/2$ doublet, with $I_3(p) = +1/2, I_3(n) = -1/2$

- **Strangeness**

  Why some baryons decay fast, while some are not?

$$\Delta^{++} \rightarrow p + \pi^+ \quad \text{Lifetime}: \sim 10^{-23} s \tag{36}$$

$$\Lambda^0 \rightarrow p + \pi^- \quad \text{Lifetime}: \sim 10^{-10} s \tag{37}$$

  Gell-Mann and Nishijima proposed **strangeness** $\mathcal{S}$, which is conserved in strong and EM interactions, but can be violated in weak interactions.

- **Strangeness**

	Δ^{++}	Λ^0	p	$\pi^{\pm}$
$\mathcal{S}$	0	**-1**	0	0

Table 2: Strangeness of particles involved.

Therefore, $\Lambda^0 \to p + \pi^-$ decay is mediated by **weak interaction** $\mathcal{S} : -1 \to 0$

- **Hypercharge**

Gell-Mann and Nishijima found :

$$Q = I_3 + \frac{Y}{2}, \quad Y = B + \mathcal{S} \tag{38}$$

The quantity Y is called **hypercharge**

The Eightfold Way

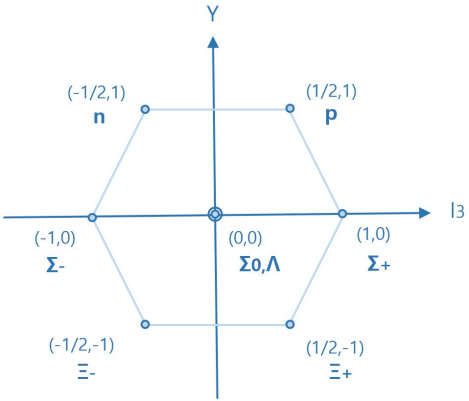


Figure 5: Arrangement of baryons by weight (I_3, Y).

- $SU(3)$ **flavor symmetry**

  In modern perspective, isospin is a symmetry of interchanging u and d quarks, since $m_u \sim m_d$

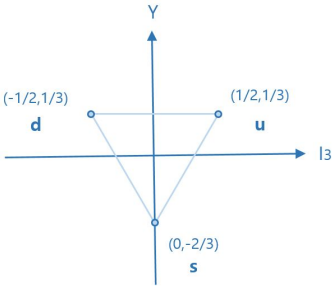


Figure 6: $SU(3)$ flavor symmetry of quarks

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Summary

- **Weights** label states by the eigenvalues of Cartan generators, while **roots** describe how these weights are connected by ladder operators.
- Every pair of opposite roots generates an $SU(2)$ subalgebra, imposing strong constraints on the geometry of the root system.
- $SU(3)$ has rank 2, with six nonzero roots and two zero weights in its adjoint representation.
- Approximate $SU(3)$ flavor symmetry organizes hadrons into multiplets, giving rise to the **Eightfold Way**.